

Linear Classifiers and the Linear SVM

D.A. Forsyth, UIUC

Linear Classifiers

- Key ideas
 - data $(\mathbf{x}_i, y_i)$
 - $\mathbf{x}_i$ are feature vectors, dimension d
 - y_i are labels, and y_i is either 1 or -1
 - query is $\mathbf{x}$
 - predicted label is:

$$\text{sign}(\mathbf{a}^T \mathbf{x} + b)$$

Question: how to choose a and b ?

- Rough answer:
 - split dataset into train/test
 - choose a and b to get good behavior on train
 - how? details?
 - evaluate on test

Question: how to choose a and b?

- Choose a and b to get good behavior on train
 - write a cost function $C(a, b)$ and find best
- The cost function:
 - generally, each training example should contribute evenly
 - average of per example costs

$$C(\mathbf{a}^T, b) = \left(\frac{1}{N}\right) \sum_{i=1}^N c(\mathbf{a}, b, \mathbf{x}_i)$$

Desirable properties $c(a, b, \mathbf{x}_i)$

- Write:

$$\gamma_i = \mathbf{a}^T \mathbf{x}_i + b$$

- sign is the predicted label
- magnitude is prop to distance to line

- We want:

- Prediction $\neq$ true label:
 - $c(a, b, \mathbf{x}_i)$ should be big. Should get bigger when magnitude is bigger, but not too fast
- Prediction == label, mag small:
 - $c(a, b, \mathbf{x}_i)$ should be non-zero, getting smaller as magnitude gets bigger
- Prediction==label, mag big:
 - $c(a, b, \mathbf{x}_i)$ should be zero

The hinge loss

$$c(\mathbf{a}, b, \mathbf{x}_i) = \max(1 - (\mathbf{a}^T \mathbf{x}_i + b) y_i, 0) = \max(1 - \gamma_i y_i, 0)$$

Feature vector for example



True label for example



The hinge loss

$$c(\mathbf{a}, b, \mathbf{x}_i) = \max(1 - (\mathbf{a}^T \mathbf{x}_i + b) y_i, 0) = \max(1 - \gamma_i y_i, 0)$$

- Notice:
 - Prediction $\neq$ true label:
 - $c(\mathbf{a}, b, \mathbf{x}_i)$ is big. Gets bigger when magnitude grows, but not too fast
 - Prediction == label, mag small:
 - $c(\mathbf{a}, b, \mathbf{x}_i)$ is non-zero, getting smaller as magnitude gets bigger
 - Prediction==label, mag big:
 - $c(\mathbf{a}, b, \mathbf{x}_i)$ is zero

